



Calibrated and recalibrated expected improvements for Bayesian optimization

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Abstract

Expected improvement (EI), a function of prediction uncertainty $\sigma(\mathbf{x})$ and improvement quantity ($\xi - \hat{y}(\mathbf{x})$), has been widely used to guide the Bayesian optimization (BO). However, the EI-based BO can get stuck in sub-optimal solutions even with a large number of samples. The previous studies attribute such sub-optimal convergence problem to the “over-exploitation” of EI. Differently, we argue that, in addition to the “over-exploitation”, EI can also get trapped in querying samples with maximum $\sigma(\mathbf{x})$ but poor objective function value $y(\mathbf{x})$. We call such issue as “over-exploration”, which can be a more challenging problem that leads to the sub-optimal convergence rate of BO. To address the issues of “over-exploration” and “over-exploitation” simultaneously, we propose to calibrate the incumbent ξ adaptively instead of fixing it as the present best solution in the EI formulation. Furthermore, we propose two calibrated versions of EI, namely calibrated EI (CEI) and recalibrated EI (REI), which combine the calibrated incumbent $\xi^{\text{Calibrated}}$ with distance constraint to enhance the local exploitation and global exploration of promising areas, respectively. After that, we integrate EI with CEI & REI to devise a novel BO algorithm named as CR-EI. Through tests on seven benchmark functions and an engineering problem of airfoil optimization, the effectiveness of CR-EI has been well demonstrated.

Keywords Expected improvement · Bayesian optimization · Gaussian process

1 Introduction

Bayesian optimization (BO) (Frazier 2018), also known as efficient global optimization (EGO) (Jones et al. 1998; Viana et al. 2013; Zhang et al. 2018), has been extensively used for solving black-box problems (Guo et al. 2018; Martinez-Cantin 2018; Min et al. 2020), such as aircraft design (Bartoli et al. 2019; Chen et al. 2019). By tradeoff between exploration (where the objective function is uncertain) and exploitation (where the objective function is expected to be desirable) in the iteration process, BO can often achieve the

optimal solution with a few number of function evaluations. Usually, a BO algorithm starts by building a Gaussian process (GP) (Williams 2006) to approximate the underlying objective function. Then, it employs an acquisition function (Shahriari et al. 2015) that incorporates information from the GP surrogate to select the most promising solution candidate to query. The algorithm repeats the above process until the termination condition is met.

The acquisition function plays a crucial role that determines the performance of BO (Shahriari et al. 2015; Guo et al. 2021). Among various acquisition functions such as probability of improvement (PI) (Viana and Haftka 2010; Ruan et al. 2020), upper-bound confidence (UBC) (Contal et al. 2013), etc., the function of expected improvement (EI) (Zhan and Xing 2020) is most popular, as it is in a simple closed form which can be efficiently optimized to yield promising solutions of the target problem (Wang and Freitas 2014).

Note that EI is a function of the prediction uncertainty $\sigma(\mathbf{x})$ and the improvement quantity $\xi - \hat{y}$, where ξ is the incumbent which is often set to be the present best solution (i.e., y_{PBS}) (Nguyen et al. 2017). Although tremendous

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successful stories have been reported (Song et al. 2016; Salemi et al. 2014; Zhaoyi et al. 2020; Li et al. 2017), the EI-based BO can get stuck in a sub-optimal solution even for a smooth function with a large number of samples (Bull 2011; Söbester et al. 2005).

The previous studies attribute the above sub-optimal convergence issue of BO to the “over-exploitation” (Chen et al. 2019) of EI. That is, EI focuses nearly all sampling efforts around the optima of the fitted GP surrogate (local exploitation), but it does not sufficiently visit other regions of the design space (global exploration) (Qin et al. 2017). To address the issue of “over-exploitation”, some previous studies propose to enhance the global exploration by augmenting the role of $\sigma(\mathbf{x})$ in the EI formulation (Ponweiser et al. 2008; Wang et al. 2017; Kleijnen 2012; Snoek et al. 2012). Alternatively, some other works propose to reduce the value of EI near the currently sampled points by changing the incumbent (Julian Berk et al. 2018). As a third option, some works propose to evaluate samples other than that with maximum EI (Zhan et al. 2017; Zhang et al. 2019), which can be also helpful to enhance the global exploration of promising areas (Mohammadi et al. 2020).

Different from the previous studies, we argue that the “over-exploitation” of EI can only be part of the reason that leads to the sub-optimal convergence problem of BO. In addition to the “over-exploitation”, EI can get trapped in querying samples with maximum prediction uncertainty $\sigma(\mathbf{x})$ but poor objective function value $y(\mathbf{x})$. We call such issue as “over-exploration”, which can be a more detrimental problem that leads to the sub-optimal convergence rate of BO under both single- and multi-fidelity contexts (Julian Berk et al. 2018; Kandasamy et al. 2017; Wu et al. 2020; Imani and Ghoreishi 2021).

More specifically, we find that the parameter $z = (\xi - \hat{y})/\sigma$ plays a crucial role that determines the performance of EI. While the “over-exploitation” may happen when $z > 0$, EI can be also trapped in “over-exploration” with $z \leq 0$. Nevertheless, the issue of “over-exploration” is rarely addressed in the existing studies (Ponweiser et al. 2008; Julian Berk et al. 2018; Zhan et al. 2017).

Based on the above findings, instead of fixing $\xi = y_{\text{PBG}}$ in the EI formulation (Jones et al. 1998), we propose to change the incumbent ξ adaptively. Moreover, we propose two calibrated versions of EI, namely calibrated EI (CEI) and recalibrated EI (REI), which appropriately adapts ξ with distance constraint to enhance the local exploitation and global exploration of promising areas, respectively.

Also note that, EI can be very efficient when it works under normal condition (Ryzhov 2016). Therefore, instead of replacing the original EI directly as those in most of the existing studies (Chen et al. 2019; Ponweiser et al. 2008), we propose to integrate EI with CEI & REI adaptively in the iteration process to achieve more effective

exploitation-exploration tradeoff. In other words, CEI & REI will be switched adaptively when there are warnings showing that the EI-based sampling is trapped in the issue of “over-exploration” or “over-exploitation”.

With the above, the main contributions of this work are summarized as follows:

- (1) Through detailed analysis, we show that the parameter $z = (\xi - \hat{y})/\sigma$ can play a crucial role to determine the performance of EI. While the so-called “over-exploitation” may happen when $z > 0$, EI can also get stuck in “over-exploration” when $z \leq 0$, which can be a more detrimental issue that leads to the sub-optimal convergence rate of BO.
- (2) Based on the above findings, we propose two calibrated versions of EI, CEI and REI, to address the “over-exploration” and “over-exploitation” issues simultaneously. Correspondingly, we propose a novel BO algorithm labeled here as CR-EI, which integrates EI with CEI & REI to achieve more effective exploitation-exploration tradeoff.
- (3) Through tests on seven benchmark functions and one engineering problem of airfoil design, and comparisons against other EI-based BO algorithms, the effectiveness of CR-EI is well demonstrated.

The remainder of this paper is organized as follows. Section 2 presents the basic knowledge of BO. And then, Sect. 3 shows the issues of EI. Thereafter, we propose two calibrated versions of EI and a novel BO algorithm labeled as CR-EI. Then, we carry out experimental studies on benchmark functions and an engineering problem in Sect. 4. Finally, we draw conclusions in Sect. 5.

2 Basics of Bayesian optimization

As Gaussian process and acquisition function constitute the key elements of Bayesian optimization, we present a brief introduction of them in this section.

2.1 Gaussian process

Gaussian process (GP) (Rasmussen 2003) is a stochastic process to approximate the underlying input-output relation, which formulates the function prediction $Y(\mathbf{x})$ as a Gaussian random variable in the function space:

$$Y(\mathbf{x}) \sim \text{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')) \quad (1)$$

where $m(\mathbf{x})$ denotes the mean function, and $k(\mathbf{x}, \mathbf{x}')$ is the covariance function between samples $\mathbf{x}$ and $\mathbf{x}'$. Usually, we take $m(\mathbf{x}) \equiv 0$ and use the popular squared exponential (SE)

covariance function (Rasmussen and Nickisch 2010; Liu et al. 2018) to describe the covariance correlations.

Given the samples $\mathbf{y}(X) = [y(\mathbf{x}_1), y(\mathbf{x}_2) \cdots y(\mathbf{x}_n)]^T$, the posterior estimates of the mean function prediction $\hat{y}$ and the corresponding mean squared error σ^2 at a point $\mathbf{x}$ can be calculated as:

$$\begin{aligned} \hat{y}(\mathbf{x}) &= \mathbf{k}(\mathbf{x}, X) (K + \sigma_n^2 \mathbf{I})^{-1} \mathbf{y} \\ \sigma^2(\mathbf{x}) &= \mathbf{k}(\mathbf{x}, \mathbf{x}) - \mathbf{k}(\mathbf{x}, X) (K + \sigma_n^2 \mathbf{I})^{-1} \mathbf{k}(X, \mathbf{x}) + \sigma_n^2 \end{aligned} \quad (2)$$

where σ_n^2 is the Gaussian noise, K is the covariance matrix over the input features of samples X , and $\mathbf{k}$ is the cross-correlation vector between $\mathbf{x}$ and X . More details of GP can refer to (Rasmussen 2003).

2.2 Acquisition function

Since the target objective function can be quite expensive to evaluate, the acquisition function acts as a surrogate that determines which sample should be queried next in Bayesian optimization. Usually, the acquisition function takes both the predicted function value $\hat{y}(\mathbf{x})$ and the prediction uncertainty $\sigma(\mathbf{x})$ into account. It aims to minimize the number of objective function calls by balancing between exploration and exploitation in the iteration process. Additionally, the acquisition functions are often required to be cheaply optimized using standard optimization packages (Frazier 2018).

Among various acquisition functions, expected improvement (EI) (Jones et al. 1998) is most frequently used. Assuming that the function prediction $Y(\mathbf{x})$ comes from a GP with the posterior estimate as $Y(\mathbf{x}) \sim N(\hat{y}(\mathbf{x}), \sigma^2(\mathbf{x}))$, the basic idea of EI is to measure how much objective improvement can be made with respect to the incumbent ξ , which can be formulated as:

$$\begin{aligned} EI(\mathbf{x}) &= \underbrace{(\xi - \hat{y}(\mathbf{x}))\Phi(z)}_{\text{exploitation}} + \underbrace{\sigma(\mathbf{x})\phi(z)}_{\text{exploration}} \\ z &= (\xi - \hat{y}(\mathbf{x})) / \sigma(\mathbf{x}) \end{aligned} \quad (3)$$

where $\Phi(\cdot)$ and $\phi(\cdot)$ denotes the standard normal cumulative and density function, respectively. And ξ is often set to be the present best solution, y_{PBS} , i.e., $\xi \equiv y_{\text{PBS}}$. In each optimization cycle, we select the next point to query by maximizing the EI, abbreviated as $\max EI$:

$$\mathbf{x}_{EI}^* = \arg \max_{\mathbf{x}} EI(\mathbf{x}) \quad (4)$$

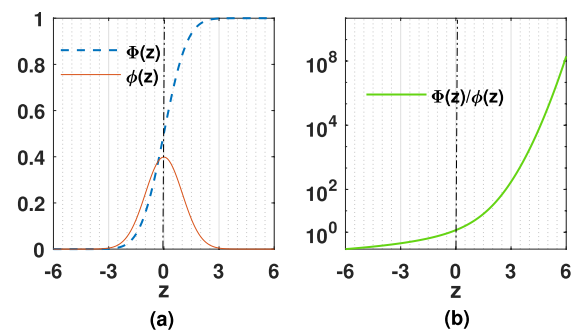


Fig. 1 Plots of a $\Phi(z)$ and $\phi(z)$ and b $\Phi(z)/\phi(z)$

3 CR-EI

In this section, we first present an analysis of EI to show the issues such as “over-exploitation” and “over-exploration”. Then, we propose two calibrated EI formulations to address the above issues. After that, we integrate the original EI with the two calibrated formulations to achieve more effective exploitation-exploration tradeoff in Bayesian optimization. The newly proposed BO algorithm is labeled as CR-EI.

3.1 Analyzing the behavior of EI

Let $F_{Y(\mathbf{x})}$ and $F'_{Y(\mathbf{x})}$ denotes the normal cumulative and density probability distribution of $Y(\mathbf{x})$, respectively, the expression of EI can be derived as:

$$\begin{aligned} EI(\mathbf{x}) &= \int_{-\infty}^{\infty} \max(y_{\text{PBS}} - Y(\mathbf{x}), 0) dF_{Y(\mathbf{x})} \\ &= \int_{-\infty}^{y_{\text{PBS}}} (y_{\text{PBS}} - \hat{y}) dF_{Y(\mathbf{x})} - \int_{-\infty}^{y_{\text{PBS}}} (Y - \hat{y}) dF_{Y(\mathbf{x})} \\ &= (y_{\text{PBS}} - \hat{y}) F_{Y(\mathbf{x})}(y_{\text{PBS}}) + \sigma F'_{Y(\mathbf{x})}(y_{\text{PBS}}) \end{aligned} \quad (5)$$

The partial derivatives of EI with respect to $\hat{y}$ and σ and the gradient of EI are:

$$\begin{aligned} \frac{\partial(EI)}{\partial \hat{y}} &= -F_{Y(\mathbf{x})}(y_{\text{PBS}}) = -\Phi\left(\frac{y_{\text{PBS}} - \hat{y}}{\sigma}\right) = -\Phi(z) \\ \frac{\partial(EI)}{\partial \sigma} &= F'_{Y(\mathbf{x})}(y_{\text{PBS}}) = \phi\left(\frac{y_{\text{PBS}} - \hat{y}}{\sigma}\right) = \phi(z) \end{aligned} \quad (6)$$

$$\begin{aligned} -\nabla EI(\mathbf{x}) &= -\left(\frac{\partial EI}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial \mathbf{x}} + \frac{\partial EI}{\partial \sigma} \frac{\partial \sigma}{\partial \mathbf{x}}\right) \\ &= \Phi(z) \frac{\partial \hat{y}}{\partial \mathbf{x}} - \phi(z) \frac{\partial \sigma}{\partial \mathbf{x}} \end{aligned} \quad (7)$$

The exploitation-exploration tradeoff effort of EI can be seen from the balance between $\partial \hat{y} / \partial \mathbf{x}$ and $\partial \sigma / \partial \mathbf{x}$ to attain

maximum EI in Eq. (7). In particular, the weights of the partial derivatives, i.e., $\Phi(z)$ and $\phi(z)$, have overwhelming effects on the exploitation-exploration tradeoff, which are depicted in Fig. 1.

When $z > 0$, the weight of $\partial\hat{y}/\partial\mathbf{x}$ (i.e., $\Phi(z)$) can be significantly greater than that of $\partial\sigma/\partial\mathbf{x}$ (i.e., $\phi(z)$). We can derive the following equation when $z > 3$:

$$\begin{aligned} -\nabla EI(\mathbf{x}|z > 3) &= \Phi(z > 3) \frac{\partial\hat{y}}{\partial\mathbf{x}} - \phi(z > 3) \frac{\partial\sigma}{\partial\mathbf{x}} \\ &\approx \Phi(z > 3) \frac{\partial\hat{y}}{\partial\mathbf{x}} \end{aligned} \quad (8)$$

That is, $\Phi(z)$ is overwhelmingly greater than $\phi(z)$ when $z > 3$ (see Fig. 1), and hence the maximization of EI is approaching to the optimal of $\hat{y}(\mathbf{x})$. In other words, EI focuses sampling near the optima of the fitted GP surrogate when z takes large positive values, and thereby it may ignore exploring other regions of the optimization space and thus get stuck in a sub-optimal solution, which is known as the issue of “over-exploitation”.

Also note that, the “over-exploitation” can only happen temporarily, as the condition $z > 0$ (i.e., $\hat{y}(\mathbf{x}_{EI}^*) < y_{\text{PBS}}$) can not be always met in the iteration process. In contrast, EI may query samples with poor $y(\mathbf{x})$ continuously and even forever when $z < 0$, as shown in the following discussion.

When $z \leq 0$, $\phi(z)$ gradually becomes greater than $\Phi(z)$, as shown in Fig. 1. In particular, we can “approximately” have the following equation with $z < -3$:

$$\begin{aligned} -\nabla EI(\mathbf{x}|z < -3) &= \Phi(z < -3) \frac{\partial\hat{y}}{\partial\mathbf{x}} - \phi(z < -3) \frac{\partial\sigma}{\partial\mathbf{x}} \\ &\approx -\phi(z < -3) \frac{\partial\sigma}{\partial\mathbf{x}} \end{aligned} \quad (9)$$

It means, as no objective improvement can be expected from the fitted GP surrogate (i.e., $z \leq 0$ with $\hat{y}(\mathbf{x}) \geq y_{\text{PBS}}$), EI turns to exploring the optimization space globally by imposing considerably greater weight on $\sigma(\mathbf{x})$. Nevertheless, $\sigma(\mathbf{x})$ has nothing to do with good or poor $y(\mathbf{x})$. Instead, the magnitude of $\sigma(\mathbf{x})$ is directly related to the density of samples around $\mathbf{x}$ (Liu et al. 2018). Hence, the performance of global exploration via $\sigma(\mathbf{x})$ can be even worse than the random search, which can be inferred from a historical view of the optimization process.

Specifically, the EI-based optimization often starts with $z > 0$. The condition $z > 0$ confines $\hat{y}(\mathbf{x}_{EI}^*)$ to be queried in promising areas with relatively better $\hat{y}(\mathbf{x})$, as shown below:

$$\begin{aligned} z &= \frac{y_{\text{PBS}} - \hat{y}(\mathbf{x})}{\sigma(\mathbf{x})} > 0 \Rightarrow \hat{y}(\mathbf{x}_{EI}^*) < y_{\text{PBS}} \\ z &= \frac{y_{\text{PBS}} - \hat{y}(\mathbf{x})}{\sigma(\mathbf{x})} \leq 0 \Rightarrow \text{No constraints on } \hat{y}(\mathbf{x}_{EI}^*) \end{aligned} \quad (10)$$

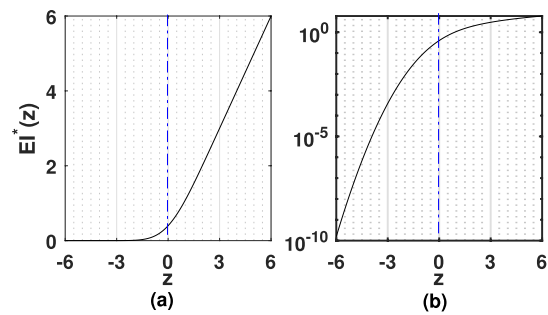


Fig. 2 Plots of $EI^*(z)$ in **a** normal scale and **b** logarithmic scale

Therefore, as the iteration goes on particularly in the stages with $z > 0$, the density of samples at regions with relatively better $y(\mathbf{x})$ becomes denser than those with poor $y(\mathbf{x})$. As a result, the sample with maximum $\sigma(\mathbf{x})$ is more likely to be located in the region with poor $y(\mathbf{x})$. However, at later stage when EI turns to exploring the optimization space globally by putting considerably greater weight on $\sigma(\mathbf{x})$, the condition $z \leq 0$ does not impose any constraints on $\hat{y}(\mathbf{x}_{EI}^*)$ (see Eq. (10)). Hence, EI is likely to query samples with maximum $\sigma(\mathbf{x})$ but poor $y(\mathbf{x})$ continuously and even forever. We would call such issue with $z \leq 0$ as “over-exploration”.

The deficiency of EI with $z \leq 0$ can also be seen from the following equation:

$$\begin{aligned} EI(\mathbf{x}) &= \sigma(\mathbf{x}) \cdot EI^*(z) \\ EI^*(z) &= z\Phi(z) + \phi(z) \end{aligned} \quad (11)$$

Accordingly we plot the curve of $EI^*(z)$ in Fig. 2. $EI^*(z)$ increases almost linearly when $z > 0$. However, it drops exponentially when $z \leq 0$. It means, when EI turns to exploring the optimization space globally with $z \leq 0$, the magnitude of EI values can be quite small, which may cause difficulty in numerical optimization. From this point of view, it should be wise to have $z > 0$ when carrying out optimization with EI (Nguyen et al. 2017).

The influence of “over-exploration” with $z \leq 0$ can be seen intuitively through the convergence history of an optimization for Hartman6 function in Fig. 3, with different indicators. Note that the Hartman6 function has six local optima, we use Latin hypercube sampling (LHS) (Helton and Davis 2003) to generate 36 samples as the initial training set to start the optimization process.

As shown in Fig. 3, after a short period of normal exploitation-exploration (marked as A1), EI gets stuck in “over-exploration” (marked as B1 to B4). And then, it takes a long period of time for EI to achieve new objective improvements (marked as A2). The “over-exploitation” issue is not clearly observed in this case, as it may happen temporarily.

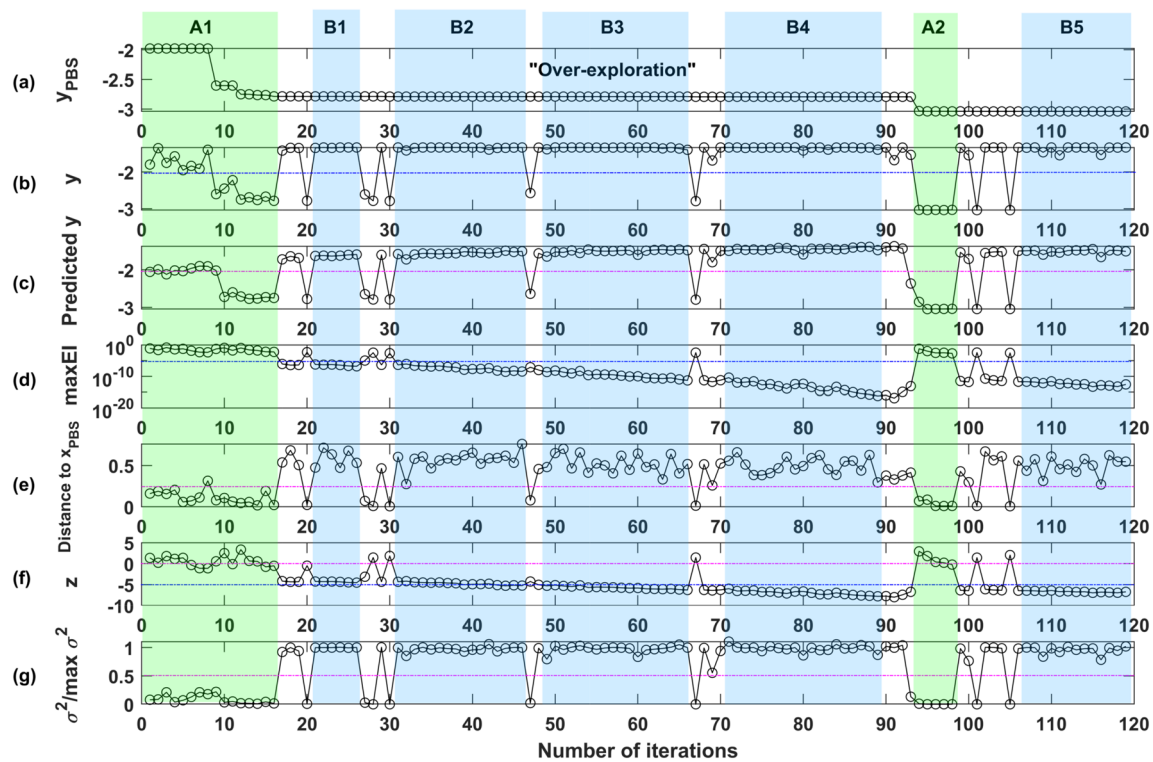


Fig. 3 Convergence behavior of EI-based sampling shown through an optimization of Hartman6 function

If we take a closer look at different periods, z is above or fluctuates around zero in the period of A1. Accordingly, the values of $\max EI$ are distributed in a normal range from 10^{-2} to 1. In contrast, when EI gets stuck in “over-exploration” as shown in the periods of B1 to B4, z fluctuates around -5, and the values of $\sigma^2 / \max \sigma^2$ are close to 1, indicating that the maximization of EI is approaching to the maximal $\sigma^2(\mathbf{x})$. Correspondingly, the magnitude of $\max EI$ ranges from 10^{-5} toward 10^{-20} , EI continuously queries samples with poor $y(\mathbf{x})$, which matches well with the analysis of Eq. (9) and (11).

Further, if we address the “over-exploration” by imposing a threshold of $\max EI$ (Nguyen et al. 2017), we may face the issue of premature convergence. That is, if we stop the algorithm in some time between B1 and B5 in this case, the optimal solution we obtained is -2.7890, which is far away from the true optimal solution of -3.0424.

To summarize, z plays a crucial role that determines the performance of EI. Compared to the “over-exploitation”, the “over-exploration” can be a more challenging problem that leads to the sub-optimal convergence rate of Bayesian optimization. Additionally, it is sensible to have $z \geq 0$ when carrying out optimization with EI; however, we may face the premature convergence issue when stopping the algorithm early with a threshold of $\max EI$.

3.2 Calibrated and recalibrated EI

To address the issues of “over-exploration” and “over-exploitation” simultaneously, we propose to change the incumbent, i.e., using $\tilde{z} = (\xi^{\text{Calibrated}} - \hat{y}(\mathbf{x})) / \sigma(\mathbf{x})$ other than $z = (y_{\text{PBS}} - \hat{y}(\mathbf{x})) / \sigma(\mathbf{x})$ in the EI formulation. Accordingly, we propose two calibrated versions of EI, namely calibrated EI (CEI) and recalibrated EI (REI), which are discussed in detail in this subsection.

3.2.1 Calibrated EI

Equation (12) shows the proposed formulation of calibrated EI, abbreviated as CEI:

$$\begin{aligned} CEI(\mathbf{x}) &= \sigma(\mathbf{x}) \cdot (\tilde{z} \Phi(\tilde{z}) + \phi(\tilde{z})) \\ \tilde{z} &= (\xi^{\text{Calibrated}} - \hat{y}(\mathbf{x})) / \sigma(\mathbf{x}) \\ \xi^{\text{Calibrated}} &> y_{\text{PBS}} \end{aligned} \quad (12)$$

Other than fixing $\xi = y_{\text{PBS}}$ as that in the original EI (see Eq. (3)), we propose to set $\xi^{\text{Calibrated}} > y_{\text{PBS}}$. Therefore, when the original EI turns to exploring the optimization space “aimlessly” with $z \leq 0$, CEI can have many points that with $z \leq 0$ to $\tilde{z} > 0$. As such, using CEI instead of EI can mitigate the risk of “over-exploration”. Moreover, it helps to enhance

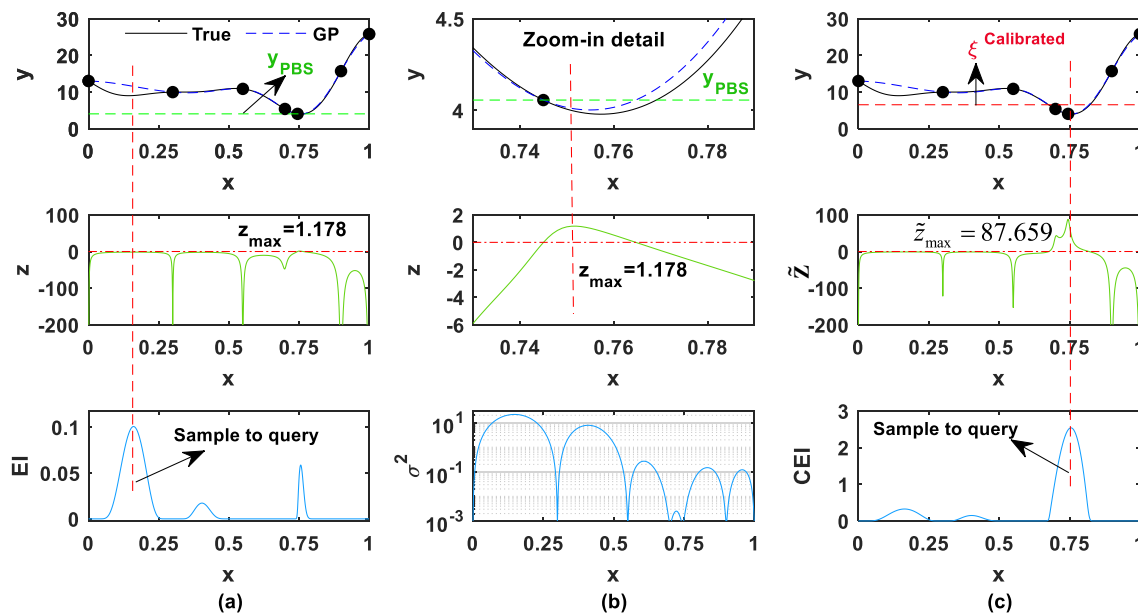


Fig. 4 Calibrated EI (CEI) to enhance the local exploitation of promising areas

the local exploitation of promising areas, as shown in the optimization for an one-dimensional toy problem in Fig. 4.

As a contrast, Fig. 4a, b show the sampling behavior and the related details of EI, where most of the z values are distributed below zero, with $z_{\max} = 1.78$. In the meantime, $\sigma(\mathbf{x})$ is quite large in some areas, making the exploitation effort as $\hat{y}(\mathbf{x}) - y_{\text{PBS}}$ be overwhelmed by the exploration effort as $\sigma(\mathbf{x})$ (see Eq. (3) and (11)). Hence, EI turns to the unnecessary exploration by querying the sample with maximum $\sigma(\mathbf{x})$ but poor $\hat{y}(\mathbf{x})$ and $y(\mathbf{x})$ as well.

Instead, using $\xi^{\text{Calibrated}} > y_{\text{PBS}}$ in CEI, the values of $\tilde{z}$ (see Fig. 4b) is large enough to overcome the overwhelming effect of $\sigma(\mathbf{x})$. Therefore, other than the “aimless” global exploration via $\sigma(\mathbf{x})$, CEI queries the sample in the area of $\tilde{z} > 0$, which gets rid of the “over-exploration” by confining the sampling area to $\hat{y}(\mathbf{x}^*) < \xi^{\text{Calibrated}}$, as shown below:

$$\tilde{z} = \frac{(\xi^{\text{Calibrated}} - \hat{y}(\mathbf{x}))}{\sigma(\mathbf{x})} > 0 \Rightarrow \hat{y}(\mathbf{x}^*) < \xi^{\text{Calibrated}} \quad (13)$$

Meanwhile, the exploitation-exploration tradeoff effort can be seen by the fact that, other than $\tilde{z}_{\max}$, CEI selects the sample near $\tilde{z}_{\max}$ to query, as shown in the bottom of Fig. 4c.

As the main purpose of CEI is to enhance the local exploitation of EI in promising areas, we also impose a distance constraint when maximizing CEI in the real optimization process, as shown below:

$$\mathbf{x}_{\text{CEI}}^* = \arg \max (\text{CEI}(\mathbf{x}) \cdot \delta_1(\|\mathbf{x} - \mathbf{x}_{\text{PBS}}\|))$$

$$\delta_1(\|\mathbf{x} - \mathbf{x}_{\text{PBS}}\|) = \begin{cases} 1 & \text{if } \varepsilon < \|\mathbf{x} - \mathbf{x}_{\text{PBS}}\| \leq d_T \\ 0 & \text{if } \|\mathbf{x} - \mathbf{x}_{\text{PBS}}\| > d_T \end{cases} \quad (14)$$

where, d_T is the distance threshold, $\mathbf{x}_{\text{PBS}}$ is the location of the present best solution. ε is a small quantity to avoid the overlapping of samples, which is often set as $\varepsilon = 0.001$. Note that, the sample candidate suggested by CEI will be queried only when it is in the small neighborhood of $\mathbf{x}_{\text{PBS}}$. The settings of d_T and $\xi^{\text{Calibrated}}$ in Eq. (12) will be discussed in Sect. 4.

3.2.2 Recalibrated EI

Although changing z to $\tilde{z}$ in CEI helps the algorithm to address the issue of “over-exploration”, it may leads to the following problem. That is, $\tilde{z}$ may take very large positive values, and hence the algorithm may be trapped in “over-exploitation” (see Eq. (8)), i.e., as $\mathbf{x}_{\text{CEI}}^*$ is continuously queried in small neighborhood of the optima of the fitted surrogate, the algorithm may ignore exploring the other areas of the optimization space, and therefore the algorithm can be trapped in a sub-optimal solution. Moreover, $\mathbf{x}_{\text{CEI}}^*$ can be fixed at $\mathbf{x}_{\text{PBS}}$, particularly when the condition $z \leq 0$ is met over the whole optimization space for the original EI.

Nevertheless, from another prospective, we can take $\mathbf{x}_{\text{PBS}}$ as a landmark to facilitate the algorithm into exploring the promising areas other than the neighborhood of $\mathbf{x}_{\text{PBS}}$, thereby guiding the algorithm toward the true global optimal

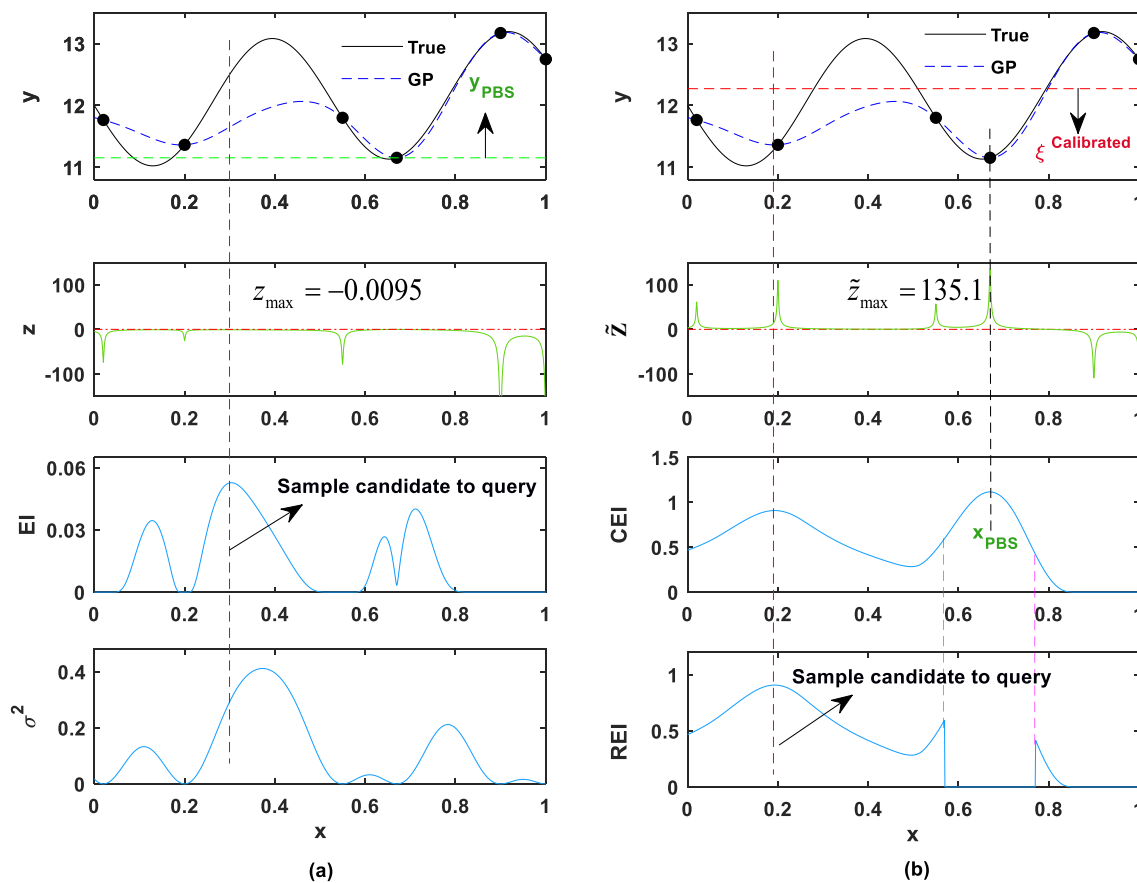


Fig. 5 Recalibrated EI (REI) to enhance the global exploration in promising areas

solution. Taking the cue, we propose to recalibrate CEI as follows:

$$REI(\mathbf{x}) = CEI(\mathbf{x}) \cdot \delta_2(\|\mathbf{x} - \mathbf{x}_{PBS}\|)$$

$$\delta_2(\|\mathbf{x} - \mathbf{x}_{PBS}\|) = \begin{cases} 0 & \text{if } \|\mathbf{x} - \mathbf{x}_{PBS}\| \leq d_T \\ 1 & \text{if } \|\mathbf{x} - \mathbf{x}_{PBS}\| > d_T \end{cases} \quad (15)$$

$$\mathbf{x}_{REI}^* = \arg \max_{\mathbf{x}} REI(\mathbf{x}) \quad (16)$$

Note that, d_T in Eq. (15) takes the same value as that shown in Eq. (14). Following this way, CEI and REI can cover the complete optimization space, when they replace EI to facilitate more effective local exploitation and global exploration, respectively. The condition as when to use CEI and REI will be discussed in detail in Sect. 3.3.

Figure 5 presents the optimization process of a multi-peak toy problem to show the sampling behavior of REI intuitively. As a contrast, Fig. 5a shows the sampling details of EI. As the values of z are below zero over the whole optimization range, EI turns to exploring the area with maximum $\sigma(\mathbf{x})$ but poor $\hat{y}(\mathbf{x})$ and $y(\mathbf{x})$. In contrast,

using $\xi^{\text{Calibrated}} < y_{PBS}$, the maximum of $\tilde{z}$ is much greater than zero. Correspondingly, the maximum of CEI is obtained at $\mathbf{x}_{PBS}$, as shown in the middle of Fig. 5b. Furthermore, by restricting the searching space to the areas other than the neighborhood of $\mathbf{x}_{PBS}$ in Eq. (15), REI selects the sample which locates in the vicinity of the real optimal solution to query. The exploitation-exploration tradeoff can also be observed in Fig. 5b, where REI selects the sample close to $\tilde{z}_{\max}$ to query.

With the above, the advantages of REI are summarized as follows. Firstly, it inherits the advantage of CEI to address the issue of “over-exploration”. Moreover, it can enhance the global exploration of the promising areas other than the vicinity of $\mathbf{x}_{PBS}$, and therefore helps to resolve the “over-exploitation” issue simultaneously.

3.3 CR-EI-based Bayesian optimization

Note that, EI can be quite efficient when it runs normally. Therefore, we propose to integrate EI with CEI & REI

adaptively to achieve improved exploitation-exploration tradeoff in Bayesian optimization. In the following paragraphs, we first discuss when to use CEI and REI in the iteration process, then we show the flowchart of the proposed Bayesian optimization algorithm, labeled as CR-EI.

Recalling that EI may get stuck in “over-exploration” at later stage of the optimization process. As a warning of it, the $\max EI$ is obtained with $z < 0$, and correspondingly $\hat{y}(\mathbf{x})$ is far away from y_{PBS} . In the meantime, as shown in Fig. 4, CEI can be quite effective to enhance the exploitation of promising areas when EI gets stuck in the issue with $z \leq 0$. Therefore, in order to decide whether to use EI or CEI to continue the optimization process automatically, we propose to use the following threshold as y_T :

$$y_T = \text{median}(\mathbf{y}(X)) \quad (17)$$

That is, the sample of EI will be queried when $\hat{y}(\mathbf{x}_{EI}^*) < y_T$, otherwise we use CEI instead. Compared to using z as a warning of the “over-exploration”, the threshold as y_T is more physically intuitive. As can be inferred from Eq. (17), the sample candidate will be queried only when the corresponding function prediction is better than half of the labeled samples. Such constraint is helpful to avoid unnecessary evaluation cost (see Fig. 4), particularly for solving continuous problems such as Lipschitz function (Hansen and Jaumard 1995).

On the other hand, REI of the proposed algorithm affects the exploration of promising areas other than the neighborhood of $\mathbf{x}_{\text{PBS}}$ (see Eq. (15)), thus enabling the algorithm to jump out of sub-optimal solutions. In other words, REI can be most effective when EI focuses on sampling near the optima of the fitted surrogate. Therefore, we propose to use REI in parallel with EI when objective improvement is observed in the iteration process, and we use an indicator named as $nBreakthrough$ to monitor the objective improvements achieved by EI.

With the above, we query one or two promising samples $\mathbf{x}_{\text{new}}^*$ in each optimization cycle, as shown below:

$$\mathbf{x}_{\text{new}}^* = \begin{cases} \{\mathbf{x}_{EI}^*\} & \text{if } nBreakthrough = 0 \\ \{\mathbf{x}_{EI}^*, \mathbf{x}_{REI}^*\} & \text{if } nBreakthrough > 0 \\ \{\mathbf{x}_{CEI}^*, \mathbf{x}_{REI}^*\} & \text{if } y(\mathbf{x}_{EI}^*) \geq y_T \end{cases} \quad (18)$$

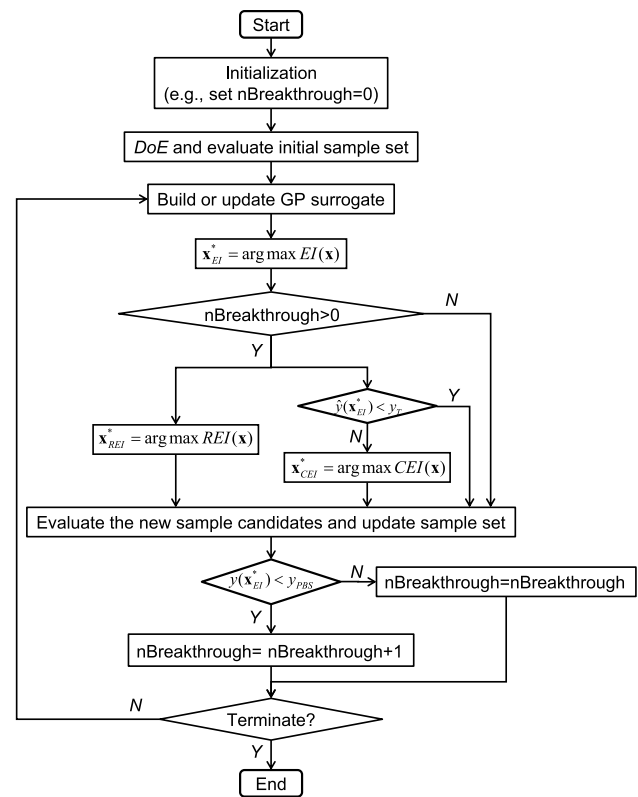


Fig. 6 Flowchart of CR-EI-based Bayesian optimization

Correspondingly, Fig. 6 shows the flowchart of CR-EI, and Algorithm 1 lists the pseudo code of CR-EI. Specifically, the algorithm starts with sampling using EI. And then, when the objective improvement of EI is observed, i.e., $nBreakthrough > 0$, REI will join in the optimization in parallel with EI, which helps to enhance the global exploration of promising areas other than the neighborhood of $\mathbf{x}_{\text{PBS}}$. On the other hand, if EI is detected to be trapped in the issue with $z \leq 0$ (indicated by $y(\mathbf{x}_{EI}^*) > y_T$), we use CEI instead of EI to enhance the local exploitation of promising areas, where the objective improvements are more likely to be attained. Thereby, the proposed CR-EI can achieve more effective exploitation-exploration tradeoff in the iteration process, and hence we can obtain the optimal solution more efficiently.

Algorithm 1 CR-EI based Bayesian optimization

Require: Target optimization problem

Ensure: Optimal solution of the target problem

```

1: Initialization, e.g., set  $nBreakThrough = 0$ .
2: Design of Experiment and evaluate the initial training
   samples
3: while the termination condition is not satisfied do
4:   Build or update GP surrogate with training samples.
5:   Carry out EI-based sample search, i.e.,  $\mathbf{x}_{EI}^* =$ 
      $\arg \max EI(\mathbf{x})$ .
6:   if  $nBreakThrough > 0$  then
7:     Conduct REI-based sample search to enhance global
       exploration, i.e.,  $\mathbf{x}_{REI}^* = \arg \max REI(\mathbf{x})$ 
8:     if  $\hat{y}(\mathbf{x}_{EI}^*) < y_T$  then
9:        $\mathbf{x}_{new}^* = \{\mathbf{x}_{REI}^*, \mathbf{x}_{EI}^*\}$ 
10:    else
11:      Conduct CEI-based sample search to enhance local
        exploitation, i.e.,
12:       $\mathbf{x}_{new}^* = \{\mathbf{x}_{REI}^*, \mathbf{x}_{CEI}^*\}$ 
13:    end if
14:  else
15:     $\mathbf{x}_{new}^* = \mathbf{x}_{EI}^*$ 
16:  end if
17:  Evaluate  $\mathbf{x}_{new}^*$  and update the training set.
18:  if  $y(\mathbf{x}_{EI}^*) < y_{PBS}$  then
19:     $nBreakThrough = nBreakThrough + 1$ 
20:  end if
21: end while
    
```

4 Experiments

In this section, we study CR-EI and compare it with a set of baselines on both benchmark functions and an engineering design problem.

4.1 Baselines for comparison

To study the effectiveness of CR-EI, we compare it with both the classical and recent EI algorithms, including EGO (Jones et al. 1998), Exploration Enhanced EI (E3I) (Julian Berk et al. 2018), Generalized EI (GEI) (Ponweiser et al. 2008) and pseudo EI (PEI) (Zhan et al. 2017).

Specifically, E3I and GEI represent the two typical efforts to address the sub-convergence rate problem of BO. More specifically, E3I proposes to use the optima of Thompson samples g^* (see Eq. (24) in Appendix 1) to replace the current best solution y_{PBS} as the incumbent in the EI formulation. As $g^* \leq y_{PBS}$, using g^* as the incumbent helps to reduce the acquisition function values near the currently sampled points, and therefore avoiding too greedily sampling around the optima of the fitted surrogate. This way, E3I is acclaimed to allow for increased early exploration while providing similar exploitation capability as EI. More details of E3I can refer to Appendix 1.

Differently, GEI proposes to adjust the weight of $\sigma(\mathbf{x})$ by adding a global factor of g (see Eq. (26) in Appendix 2).

Further, inspired by the strategy of simulated annealing (van Laarhoven and Aarts 1987), g is updated dynamically to enhance the local exploitation and global exploration, respectively, at different stages of the optimization process. We strictly follow the settings of g in Ponweiser et al. (2008) to conduct the empirical studies in this section, and more details of GEI can be seen in Appendix 2.

PEI is originally proposed to enable parallel optimization (Haftka et al. 2016). In addition to evaluating the sample of $maxEI$, PEI also queries samples other than $maxEI$ in each iteration, which is achieved by multiplying a influential function (see Eq. (27) in Appendix 3) to the original EI formulation. As shown in the subsequent study (Mohammadi et al. 2020), querying the samples other than $maxEI$ can help to enhance the exploration performance of EI-based BO. In the following tests, we follow the settings in Zhan et al. (2017) and query two samples in each iteration.

Note that the above mentioned algorithms are proposed mainly to address the “over-exploitation” of EI, thus resolving the sub-optimal convergence problem of BO. Different from the above works, we argue that the parameter $z = (\xi - \hat{y}(\mathbf{x})) / \sigma(\mathbf{x})$ plays a crucial role to determine the performance of EI. While the “over-exploitation” may happen when $z > 0$, EI can also suffer from the “over-exploration” issue when $z \leq 0$ (see Sect. 3). Hence, we propose CR-EI to address the issues of “over-exploration” and “over-exploitation” simultaneously. Also note that there are two parameters in CR-EI, i.e., d_T and $\xi^{Calibrated}$ as shown in Eqs. (12) and (15), which are set as follows:

$$d_T = \alpha \sqrt{dim} \quad (19)$$

$$\xi^{Calibrated} = \begin{cases} (1 + \beta)y_{PBS} & \text{if } y_{PBS} > 0 \\ \min \mathbf{y}(X) \text{ other than } y_{PBS} & \text{if } y_{PBS} = 0 \\ (1 - \beta)y_{PBS} & \text{if } y_{PBS} < 0 \end{cases} \quad (20)$$

where dim denotes the dimension of design space. By default, we set $\alpha = 0.1$ and $\beta = 0.1$ in the following tests. The sensitivities of α and β on algorithm performance are discussed in Sect. 4.5.

4.2 Experimental setup

For fair comparison, we use the GPML toolbox (Rasmussen and Nickisch 2010) to build a uniform test bed for CR-EI and the compared algorithms. The number of initial training samples is set as six times of the problem dimension. Since the best solution of benchmark functions are known, we use the regret (Wang and Jegelka 2017) to measure the best solution obtained thus far, i.e., $R_t = f_{\min}^{(t)} - \min_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x})$, wherein $\min_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x})$ is the real optimal solution of the benchmarks, and $f_{\min}^{(t)}$ denotes the best solution at the t^{th} iteration.

On the other hand, we show the best objective function value directly for the engineering problem due to the unknown optimum.

In addition, two conditions are used to terminate the compared algorithms, as shown below:

- (1) $n_{\text{total}} \leq n_T$, i.e., the total number of sample evaluations n_{total} for each compared algorithm should be not greater than the sample budget n_T . In general, n_T is set as 40 times of the problem dimension for the benchmark tests. As exceptions, n_T is set to be 500 for the Shekel4 function which has many local optima, and n_T is set to be 250 for the 10-dimensional Trid function with convex objective landscape.
- (2) $t_{\text{total}} \leq t_T$, i.e., the total optimization cycles t_{total} should be not greater than a threshold t_T .

Note that, the termination condition $t_{\text{total}} \leq t_T$ is mainly used to address the following issue. That is, in order to avoid the overlapping of samples and thus resolving the ill-conditioning of the correlation matrix, a distance constraint $\min\|\mathbf{x} - \mathbf{x}_i\| > \varepsilon$ is imposed, where $\mathbf{x}_i$ is one of the training samples, and we set $\varepsilon = 0.001$ by default. However, when the algorithm arrives at an optimum, the constraint $\min\|\mathbf{x} - \mathbf{x}_i\| > \varepsilon$ can prevent the algorithm from querying more samples at later stage of the iteration process, and then the algorithm may get stuck in infinite loop but cannot meet the termination condition $n_{\text{total}} \leq n_T$. Hence, we propose to stop the algorithm when either the condition (1) or (2) is met.

4.3 Tests on benchmark functions

Seven benchmark functions of different types are selected from Surjanovic and Bingham (2021). Specifically, the Sasena and Hartman functions are popular non-convex problems frequently used as benchmarks (Satria and Shimoyama 2019; Chaudhuri et al. 2013). The Camel6 and Shekel4 functions are challenging problems which have many local optima. In addition, although the 10-dimensional Trid function has no local minimum except the global one, the function values of Trid change in several orders of magnitude, and the function dimension is relatively high. Therefore, the optimal solution of Trid cannot be easily obtained.

Figures 7, 8, 9, 10, 11, 12 and 13 show the optimization results of CR-EI and the other baseline algorithms, where the shaded areas in Figs. 7a, 8a, 9a, 10a, 11a, 12a and 13a show the variations of the optimization results of different algorithms over 20 runs. The boxplots in Figs. 7b, 8b, 9b, 10b, 11b, 12b and 13b exhibit the medians and the distributions of the final optimal solutions. Except the Trid10 function, CR-EI achieves the best optimal solutions with faster convergence rates for the remaining six benchmarks.

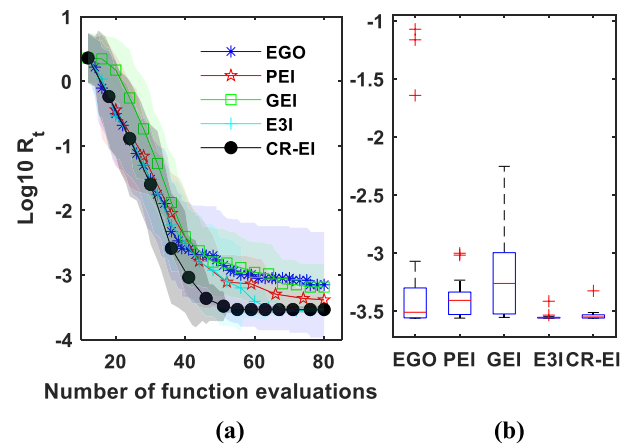


Fig. 7 Tests on Sasena function, **a** convergence history, **b** boxplot of optimal solutions

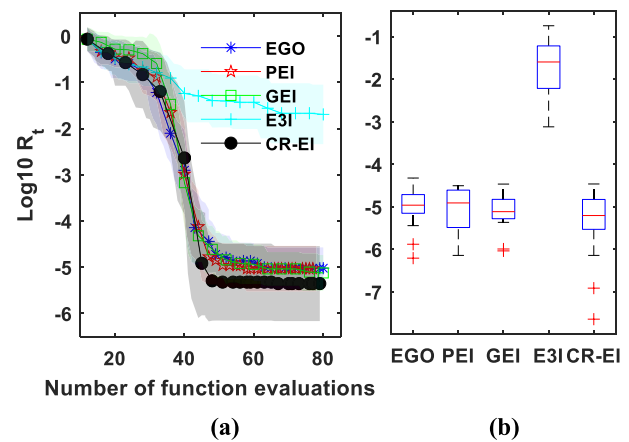


Fig. 8 Tests on Camel6 function, **a** convergence history, **b** boxplot of optimal solutions

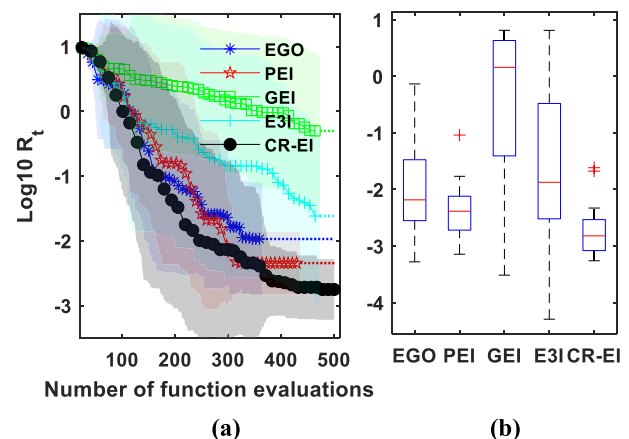


Fig. 9 Tests on Shekel4 function, **a** convergence history, **b** boxplot of optimal solutions

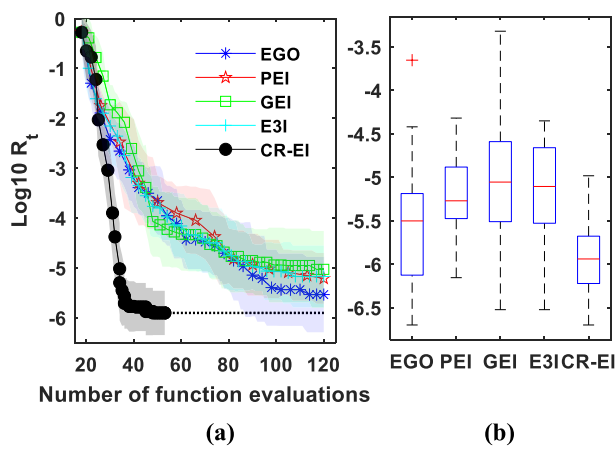


Fig. 10 Tests on Hartman3 function, **a** convergence history, **b** boxplot of optimal solutions

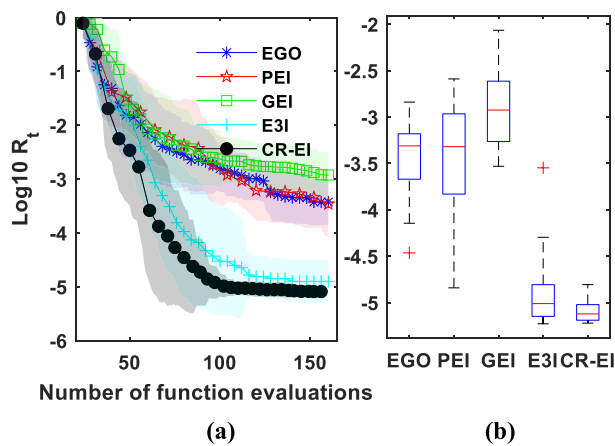


Fig. 11 Tests on Hartman4 function, **a** convergence history, **b** boxplot of optimal solutions

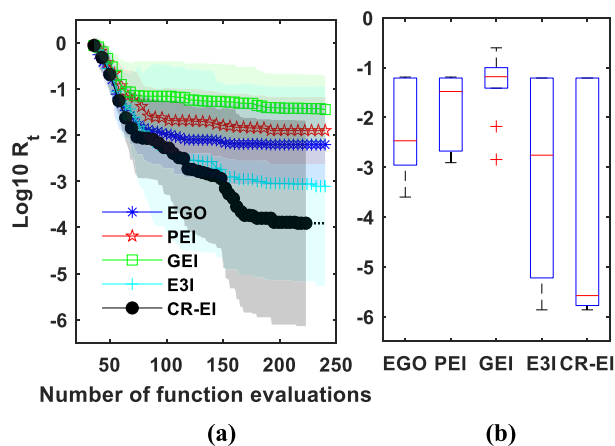


Fig. 12 Tests on Hartman6 function, **a** convergence history, **b** boxplot of optimal solutions

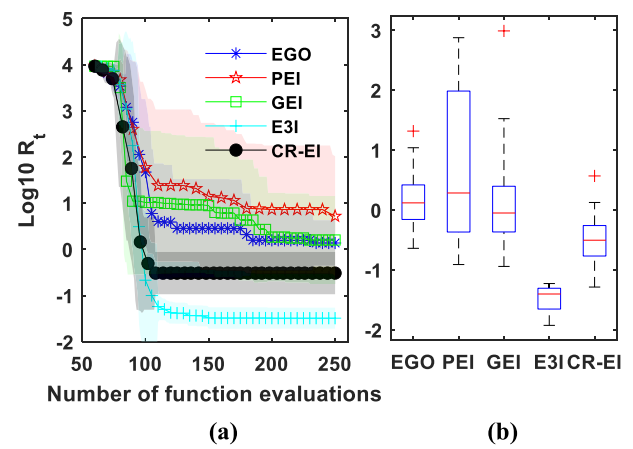


Fig. 13 Tests on Trid10 function, **a** convergence history, **b** boxplot of optimal solutions

Compared to the optimization results of EGO, E3I achieves relatively better results when optimizing Trid10, Sasena and the Hartman functions. However, the results of E3I are very poor when optimizing Camel6 and Shekel4. Note that Trid10 is a bowl-shaped function with only one global optimal solution (Surjanovic and Bingham 2021). Nevertheless, the objective landscape of Trid10 cannot be easily fitted with limited number of samples, as the dimension of Trid10 is relatively high. As the iteration goes on, the magnitude of $\sigma(\mathbf{x})$ decays rapidly, and the optimization with EI is likely to focus sampling around the optima of the fitted surrogate with large z (i.e., “over-exploitation”). After that, the algorithm can be soon trapped in querying samples with relatively larger $\sigma(\mathbf{x})$ but poor $y(\mathbf{x})$ (i.e., “over-exploration”), when no objective improvements can be attained with $z \leq 0$. Therefore, EGO often gets stuck in a sub-optimal solution, as seen in Fig. 13. In contrast, using the optima of Thompson samples g^* (see Eq. (24) in Appendix 1) instead of y_{PBS} in E3I can be quite effective to reduce the large positive values of z . Recalling the discussion with Eq. (8), reducing the positive values of z helps to prevent too greedily sampling around the optima of the fitted surrogate. Thereby, E3I is able to achieve better objective improvements in promising areas other than the small neighborhood of $\mathbf{x}_{\text{PBS}}$, which in turn helps E3I to continuously conduct effective exploitation-exploration tradeoff with $z > 0$. Hence, E3I achieves significantly better results than EGO.

Nevertheless, $z \leq 0$ can take place many times when optimizing the Camel6 and Shekel4 functions, where many local optima are located. Then, the algorithms using either EI or E3I as the acquisition function can be easily trapped in a local solution of the bumpy objective landscape, and no improvement can be observed, resulting in $z \leq 0$, i.e., $\hat{y}(\mathbf{x}) \geq y_{\text{PBS}}$. More importantly, using g^* instead of y_{PBS} as the incumbent in E3I will increase the absolute value of $z \leq 0$,

and therefore enhancing the overwhelming effect of $\sigma(\mathbf{x})$, as discussed with Eq. (9). Accordingly, the issue of “over-exploration” can be more prominent for the optimization with the E3I acquisition function. As a consequence, E3I obtains poor results when optimizing the bumpy and multi-peak functions such as Camel6 and Shekel4.

By using large global factor g to enhance exploration of the optimization space (see Appendix 2), GEI achieves relatively better results when optimizing the two-dimensional functions such as Sasena and Camel6, but the convergence rates of GEI are slower at the early stage. In the meantime, GEI performs poorly on Shekel4 and the other benchmark functions of higher dimension. The underlying reasons can be explained as follows. Firstly, the global factor g , which intends to achieve a dynamic balance in between exploitation and exploration, is set empirically (Ponweiser et al. 2008). Such empirical settings may not adapt well to the real objective landscape of higher-dimensional problems. Secondly and more importantly, *adjusting the weight of $\sigma(\mathbf{x})$ by g cannot change the value of $z \leq 0$ to address the “over-exploration” issue*. Moreover, the iterations with empirical settings of g may even make the algorithm get into the situation with $z \leq 0$ more easily, resulting in worse results than those of EGO.

By querying the sample of $\max EI$ and the one away from $\max EI$ at each optimization cycle, the performance of PEI is relatively robust. Also note that, although the additional sample other than $\max EI$ can help the algorithm to escape from a local solution, it cannot directly address the issues of “over-exploration” and “over-exploitation”. Moreover, it requires additional sample costs in each optimization cycle. Hence, within a fixed sample budget, PEI occasionally achieves better results than EGO, as seen in Figs. 7, 8, 9, 10, 11, 12 and 13.

Different from the compared baselines, we propose to address the issues of “over-exploitation” and “over-exploration” simultaneously. In doing so, we propose to replace $z \leq 0$ by $\tilde{z} > 0$ in CR-EI, particularly when there are warnings showing that EI-based sampling is trapped in the issue of “over-exploration” (see Eq. (18)). Moreover, the exploitation and exploration of promising areas are enhanced adaptively by integrating EI with CEI & REI in each iteration. Hence, CR-EI can achieve more effective exploitation-exploration tradeoff, and therefore attaining significantly better results with much faster convergence rates, as shown in Figs. 7 to 12. Besides, CR-EI even stops earlier than the other compared algorithms in Figs. 10, as it soon arrives at the optimal solution to meet the termination condition.

Table 1 summarizes the optimization results of compared algorithms on benchmark tests over 20 runs. The success of an algorithm is defined as $(y_{\min} - f_{\min})/f_{\min} < 1\%$, where $f_{\min}$ denotes the real optimal solution of a benchmark function, and $y_{\min}$ is the final solution attained by an algorithm.

Additionally, the algorithm that achieves the best results is marked in bold. Obviously, our proposed CR-EI achieves the best performance in most cases.

In particular, the standard deviation of CR-EI is larger than the compared algorithms when optimizing Hartman6. The reason behind can be explained as follows. That is, Hartman6 has several local optimal solutions, the values of which are close to each other. Specifically, two of these local optimal solutions are -3.0424 and -2.9810 . CR-EI attains the final solution at -3.0424 in 13 times out of 20 runs, and arrives at -2.9810 by 6 times. Meanwhile, CR-EI achieves the final solution as -3.0400 in one test. Similar results are observed for E3I, which arrives at or approaches to -3.0424 by 12 times, and arrives at -2.9810 by eight times. In contrast, the final solutions of the other compared algorithms are randomly distributed in the range of $[-3.042, -2.978]$, as they always converge at a local solution. Hence, the standard deviation of the optimization results of CR-EI looks greater than those of EGO, PEI and GEI. Nevertheless, the median and success rate of CR-EI are much better as shown in Table 1, which confirms the superior performance of CR-EI.

4.4 The airfoil design problem

To further show the effectiveness of CR-EI, we test it on an airfoil optimization problem.

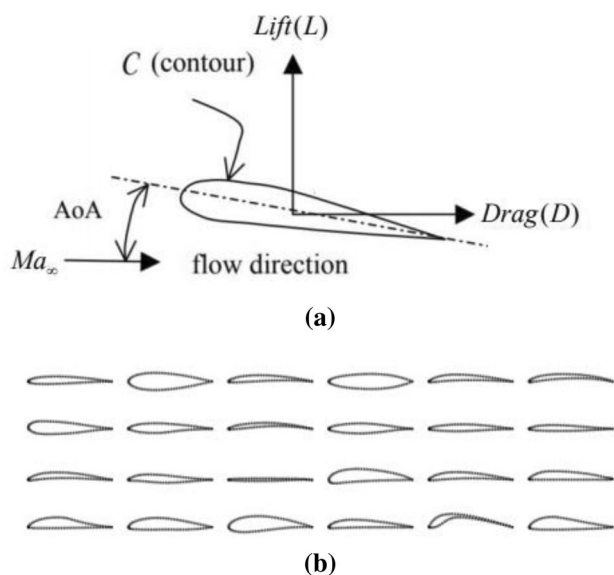
4.4.1 Problem description

Airfoil design plays a fundamental role in aircraft design (Eppler 2012). Generally, as depicted by Fig. 14a, the forces acting on the airfoil can be decomposed into two components as the drag (D) and the lift (L). The drag and lift and their ratio L/D as well are functions of the airfoil contour. In the meantime, the airfoil contours can be in various shapes according to different working conditions determined by the aerodynamic parameters such as the Mach number (Ma_{∞}), the Reynolds number (Re) and the angle of attack (AoA), as shown in Fig. 14b.

The optimization of an airfoil often starts by determining the operating condition and the objective function. Thereafter, the variation space of airfoil contours is parameterized to carry out the airfoil optimization. In this study, we follow the settings in Chen et al. (2019) for the optimization of a low-speed airfoil. The optimization objective is to maximize the lift to drag ratio, i.e., L/D , of a low-speed airfoil. The XFOIL (Drela 1989) is used to compute L/D , with the operating conditions set as $Re = 1.8 \times 10^6$, $Ma = 0.01$ and $AoA = 0$ deg.

Table 1 Optimization results of compared algorithms on seven benchmark functions

Function	Indicators	EGO	PEI	GEI	E3I	CR-EI
Sesena	Mean	9.09×10^{-3}	4.43×10^{-4}	9.89×10^{-4}	2.83×10^{-4}	2.93×10^{-3}
	Standard deviation	2.37×10^{-2}	2.03×10^{-4}	1.23×10^{-3}	2.40×10^{-5}	4.32×10^{-5}
	Median	3.08×10^{-4}	3.90×10^{-4}	5.47×10^{-4}	2.77×10^{-4}	2.81×10^{-4}
	Success rate	20/20	20/20	20/20	20/20	20/20
Camel6	Mean	1.39×10^{-5}	1.46×10^{-5}	1.02×10^{-5}	4.34×10^{-2}	1.03×10^{-5}
	Standard deviation	1.12×10^{-5}	1.11×10^{-5}	7.78×10^{-6}	4.71×10^{-2}	1.02×10^{-5}
	Median	1.10×10^{-5}	1.25×10^{-5}	7.72×10^{-6}	2.60×10^{-2}	6.32×10^{-6}
	Success rate	20/20	20/20	20/20	9/20	20/20
Shekel4	Mean	8.58×10^{-2}	9.74×10^{-3}	2.31×10^0	6.18×10^{-1}	3.77×10^{-3}
	Standard deviation	2.01×10^{-1}	1.98×10^{-2}	2.53×10^0	1.50×10^0	6.59×10^{-3}
	Median	6.55×10^{-3}	4.12×10^{-3}	1.44×10^0	1.34×10^{-2}	1.52×10^{-3}
	Success rate	17/20	20/20	6/20	14/20	20/20
Hartman3	Mean	1.66×10^{-5}	1.12×10^{-5}	4.23×10^{-5}	1.34×10^{-5}	2.18×10^{-6}
	Standard deviation	4.91×10^{-5}	1.28×10^{-5}	1.07×10^{-4}	1.32×10^{-5}	2.93×10^{-6}
	Median	3.15×10^{-6}	5.55×10^{-6}	8.95×10^{-6}	7.85×10^{-6}	1.15×10^{-6}
	Success rate	20/20	20/20	20/20	20/20	20/20
Hartman4	Mean	5.36×10^{-4}	6.82×10^{-4}	1.88×10^{-3}	2.63×10^{-5}	8.40×10^{-6}
	Standard deviation	4.04×10^{-4}	6.96×10^{-4}	2.10×10^{-3}	6.12×10^{-5}	2.42×10^{-6}
	Median	4.88×10^{-4}	4.78×10^{-4}	1.19×10^{-3}	9.75×10^{-6}	7.55×10^{-6}
	Success rate	20/20	20/20	20/20	20/20	20/20
Hartman6	Mean	2.20×10^{-2}	3.29×10^{-2}	1.06×10^{-1}	2.74×10^{-2}	2.08×10^{-2}
	Standard deviation	2.94×10^{-2}	3.00×10^{-2}	1.52×10^{-1}	3.13×10^{-2}	2.96×10^{-2}
	Median	2.35×10^{-3}	3.96×10^{-2}	6.47×10^{-2}	7.58×10^{-4}	6.31×10^{-6}
	Success rate	14/20	10/20	6/20	12/20	14/20
Trid10	Mean	2.85×10^0	1.01×10^2	8.05×10^1	3.64×10^{-2}	5.47×10^{-1}
	Standard deviation	4.83×10^0	2.08×10^2	3.43×10^2	1.59×10^{-2}	8.10×10^{-1}
	Median	1.32×10^0	2.05×10^0	1.20×10^0	3.98×10^{-2}	3.15×10^{-1}
	Success rate	14/20	10/20	13/20	20/20	20/20

**Fig. 14** Illustration of **a** the airfoil contour and the forces acting on it, and **b** the airfoil contours in UIUC database

We use the generative adversarial net (GAN) (Goodfellow et al. 2014) with UIUC database¹ to parameterize the optimization space of airfoil contours, as shown in Fig. 15. Note that, GAN often consists of two components as the generator and the discriminator. The generator is trained to synthesize contours that follow the characteristics of the real airfoils, while the discriminator guides the evolution of generator by distinguishing the synthesized airfoil from the real one in the training process. More details of GAN can refer to Goodfellow et al. (2014); Li et al. (2020). After training, the generator can synthesize airfoils of various shapes by varying the input $\mathbf{x}$.

For the optimization of a low-speed airfoil in this study, the dimension of $\mathbf{x}$ is set as 13. The 13-dimensional optimization task can thus be formulated as:

$$\min f(\mathbf{x}) = \min L/D(\mathbf{x}) \quad (21)$$

with $Ma = 0.01$, $Re = 1.8 \times 10^6$, $AoA = 0deg$.

¹ <https://m-selig.ae.illinois.edu/ads.html>.

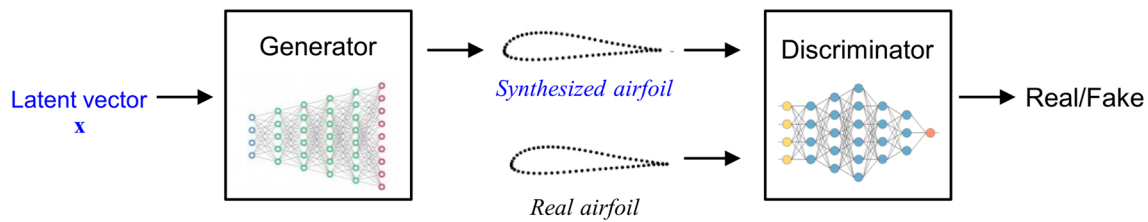


Fig. 15 Airfoil parameterization using generative adversarial net (GAN)

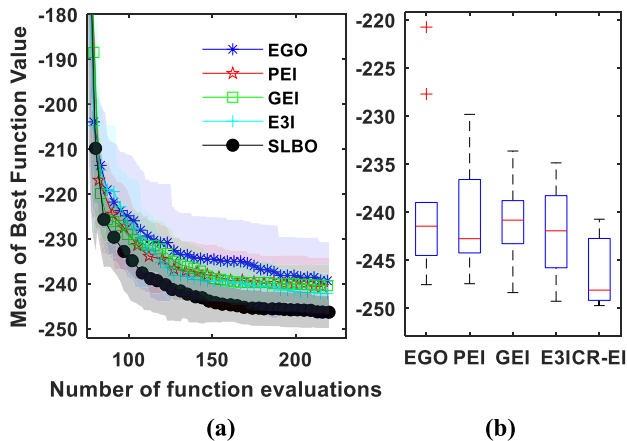


Fig. 16 Tests on airfoil design, **a** convergence history, **b** boxplot of optimal solutions

Table 2 Optimization results of compared algorithms on airfoil design

	EGO	PEI	GEI	E3I	CR-EI
Performance Indicators					
Mean $\pm$ SD	-239.34 ± 8.60	-240.66 ± 6.41	-240.70 ± 4.57	-242.20 ± 4.48	-246.34 ± 3.56
Median	-241.44	-242.74	-240.82	-241.92	-248.09

4.4.2 Results and discussion

Figure 16 shows the optimization results for the low-speed airfoil design over 10 runs, and Table 2 summarizes the statistical results of compared algorithms, where the algorithm that achieves the best results is marked in bold. Similar to the results shown in benchmark tests of toy problems, CR-EI achieves better optimization results than the compared algorithms. Moreover, the convergence rate of CR-EI is also significantly faster. The better results of CR-EI can be attributed to the following reasons. That is, we adjust the incumbent ξ in EI to have $\tilde{z} > 0$ in most periods of the optimization process, which helps to address the issue of “over-exploration”. Moreover, the local exploitation and global exploitation of promising areas are also enhanced adaptively by integrating CEI & REI

with EI. Consequently, CR-EI achieves a more effective exploitation-exploration tradeoff in the iteration process.

4.5 Sensitivity analysis of algorithm parameters

As shown in Eqs. (19) and (20), there are two parameters, i.e., the distance parameter α and the incumbent parameter β , can be tuned in the proposed CR-EI. We discuss the influences of these parameters in the following paragraphs, by testing on two benchmark functions.

4.5.1 Influence of distance parameter

Figures 17 and 18 show the optimization results by varying the distance parameter α while fixing $\beta = 0.1$. The results of CR-EI with different α are consistently better than those

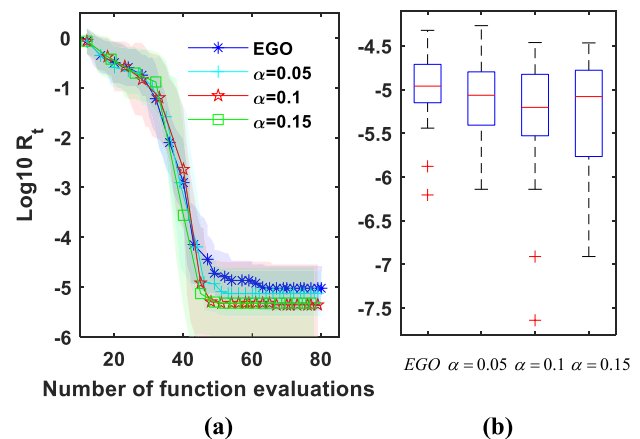


Fig. 17 Influence of distance parameter α on algorithm performance, by testing on Camel6 function

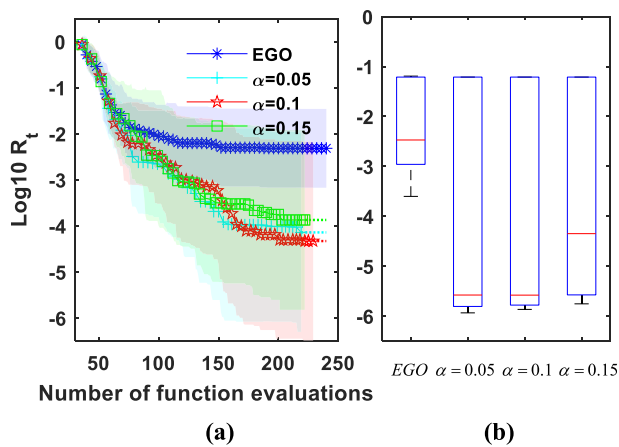


Fig. 18 Influence of distance parameter α on algorithm performance, by testing on Hartman6 function

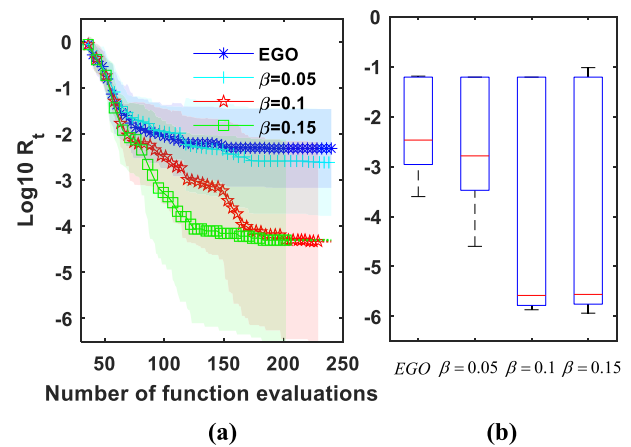


Fig. 20 Influence of incumbent parameter β on algorithm performance, by testing on Hartman6 function

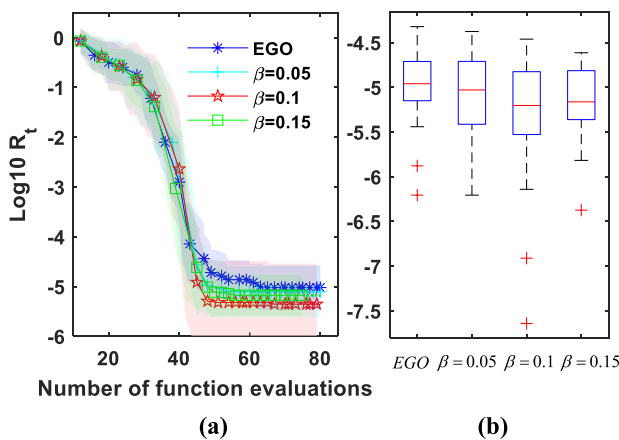


Fig. 19 Influence of incumbent parameter β on algorithm performance, by testing on Camel6 function

of EGO. Recalling the expression of REI in Eq. (15) and Eq. (19), the distance parameter α facilitates the algorithm into exploring promising areas other than the vicinity of the present best solution $\mathbf{x}_{\text{PBG}}$. Such REI-based sampling can be quite effective to help the algorithm jump out of sub-optimal solutions, making the performance of CR-EI consistently better than that of EGO.

On the other hand, the performance of CR-EI varies slightly with different α values. Specifically, while helping the algorithm jump out of sub-optimal solutions, too large α can also lead the algorithm to search the less “promising” areas with greater probability. In other words, there can be a balance when setting α to enhance the global searching performance of EI. Hence, CR-EI with moderate α (i.e., $\alpha = 0.1$ in our testing cases) achieves the

best solutions, as shown in the testing of both Camel6 and Hartman6 functions. In overall, the algorithm performance is not that sensitive to the change of the distance parameter α .

4.5.2 Influence of incumbent parameter

Figures 19 and 20 show the optimization results by varying the incumbent parameter β , where we fix $\alpha = 0.1$. Compared to the distance parameter α , the performance of CR-EI is more sensitive to the variation of β . As shown in Figs. 19a and 20a, the convergence rate of CR-EI becomes faster with the increase of β . In the meantime, CR-EI with moderate β (i.e., $\beta = 0.1$ in our testing cases) achieves the best optimal solutions at the end of the optimization process.

Looking back to Eqs. (12) and (20), the incumbent parameter β measures to what extent we can have $\tilde{z} > 0$ when $z \leq 0$. It means, the relatively larger β can help the algorithm to be more resistant to the “over-exploration” issue with $z \leq 0$. Therefore, CR-EI with larger β can facilitate effective exploitation-exploration tradeoff to achieve even faster convergence rate. In the meantime, as shown in Eq. (13), β also confines the promising sampling area to $y(\mathbf{x}^*) < \xi^{\text{Calibrated}}$. The relative larger β also means a larger area to explore. Under extreme conditions, the searching area can even cover the whole optimization space when β is set too large. In other words, there can be a balance in selecting β to achieve the optimal exploitation-exploration tradeoff in the iteration process. Hence, CR-EI with $\beta = 0.1$ achieves the best optimization results in our testing cases.

5 Conclusions

In this paper, we propose a novel Bayesian optimization (BO) algorithm named as CR-EI. Different from the previous studies which resort to adjusting $\sigma(\mathbf{x})$ to address the sub-convergence rate problem of EI-based BO, we propose to adapt the incumbent ξ in the EI formulation to address it. This is based on our findings that, the parameter $z = (\xi - \hat{y}(\mathbf{x})) / \sigma(\mathbf{x})$ plays a crucial role that determines the performance of EI. While adjusting $\sigma(\mathbf{x})$ can be helpful to address the issue of “over-exploitation” when $z > 0$, the EI-based BO can also suffer from the “over-exploration” issue when $z \leq 0$, which can be a more challenging problem that leads to the sub-convergence rate of BO.

More specifically, in order to resolve the issues of “over-exploration” and “over-exploitation” simultaneously, we propose to use $\xi^{\text{Calibrated}} > y_{\text{PBS}}$ instead of $\xi \equiv y_{\text{PBS}}$ in the EI formulation. Moreover, we propose two calibrated EI formulations, CEI and REI, which combine $\xi^{\text{Calibrated}}$ with distance constraint to enhance the local exploitation and global exploration of promising areas, respectively. Furthermore, we propose CR-EI that integrates EI with CEI&REI in the iteration process. The effectiveness of CR-EI has been well demonstrated through experimental studies on benchmark functions and an airfoil engineering optimization problem.

Appendix 1: brief introduction of E3I

According to Bochner’s theorem (Cooke and Bochner 2005), a shift invariant GP kernel, such as the square exponential (SE) kernel used in this paper, can be approximated by a Fourier function as:

$$k(\mathbf{x}_i, \mathbf{x}_j) \approx \phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j) \quad (22)$$

$$\phi(\mathbf{x}) = \sqrt{2\beta/V} \{ \cos(W\mathbf{x} + b), \sin(W\mathbf{x} + b) \}$$

where, β is the normalizing factor of a Fourier dual, W is the matrix of random weight $\mathbf{w}_i$, i.e., $W = [\mathbf{w}_1, \dots, \mathbf{w}_V]$, and b is a constant vector that follows a uniform distribution as $b \sim U[0, 2\pi]$. Given the samples $\mathbf{y} = [y(\mathbf{x}_1), y(\mathbf{x}_2) \dots y(\mathbf{x}_n)]^T$ and let $\Phi = [\phi(\mathbf{x}_1), \dots, \phi(\mathbf{x}_n)]^T$, the GP mean prediction $g(\mathbf{x})$ of the black-box function $f(\mathbf{x})$ can be expressed as:

$$f(\mathbf{x}) \approx g(\mathbf{x}) = \phi(\mathbf{x})^T (\Phi\Phi^T + \sigma^2 I)^{-1} \Phi^T \mathbf{y} \quad (23)$$

In particular, $g(\mathbf{x})$ can be viewed as a complete estimate of $f(\mathbf{x})$ when the number of weights V in the kernel function is sufficiently large. Then, full random approximations of $f(\mathbf{x})$ can be generated using Eq. (23). Hence, E3I (Julian Berk et al. 2018) is proposed which makes use of the random approximations to enhance the exploration performance of EI with Thompson sampling (Russo et al. 2017).

Specifically, instead of using the current best solution y_{PBS} as the incumbent in the EI formulation (see Eq. (11)), the optima of Thompson samples g_m^* that generated with Eq. (23) are used as the incumbent in E3I:

$$z_m^T = (g_m^* - \hat{y}(\mathbf{x})) / \sigma(\mathbf{x}) \quad (24)$$

The process for generating and finding the optima of Thompson samples is outlined in Algorithm 2. Further, considering the variations between Thompson samples, the mean of several Thompson samples is used to calculate the E3I acquisition function at an unknown point $\mathbf{x}$, as shown below:

$$E3I(\mathbf{x}) = \frac{1}{M} \sum_{m=1}^M E[I(\mathbf{x}, g_m^*)] \quad (25)$$

$$= \frac{\sigma(\mathbf{x})}{M} \left[\sum_{m=1}^M z_m^T \Phi(z_m^T) + \phi(z_m^T) \right]$$

The capability of E3I to improve the performance of EI lies in the following property of Thompson sample functions. That is, for a minimization problem, the optima of Thompson samples g^* are not greater than the current best solution y_{PBS} , i.e., $g^* \leq y_{\text{PBS}}$ (Julian Berk et al. 2018). Then, according to the analysis with Eq. (8), the smaller incumbent using g^* helps to lower the large positive values of z , and therefore lowering the weight of $\partial \hat{y} / \partial \mathbf{x}$ to avoid too greedily sampling around the optima of fitted surrogate. From this end, E3I can help to address the issue of “over-exploitation” and thus improving the performance of EI.

However, when the situation with $z \leq 0$ happens, using g^* instead of y_{PBS} will increase the absolute value of z , and thus enhancing the overwhelming effect of $\sigma(\mathbf{x})$, as discussed with Eq. (9). Accordingly, the algorithm using E3I as the acquisition function can be more likely to get stuck in querying samples with maximum $\sigma(\mathbf{x})$ but poor $y(\mathbf{x})$ when $z \leq 0$. In other words, E3I may even worsen the situation of “over-exploration” when it occurs with $z \leq 0$.

Table 3 The scheme to set the g value according to the number of iterations (Ponweiser et al. 2008)

Iteration	g value
1–4	20
5–9	10
10–19	5
20–24	2
25–34	1
≥ 35	0

Algorithm 2 Thompson Sampling

Require: $D_{t-1} = \{\mathbf{x}_i, y_i\}_{i=1}^{t-1}$, # random feature dimension, V , # Thompson samples, M

- 1: **for** $m=1$ to M **do**
- 2: Randomly generate $b \sim U[0, 2\pi]$ and V weights, $w_i \sim N(0, I_{d \times d}) \forall i = 1 \dots V$
- 3: Let $W = [w_1, \dots, w_V] \in R^{V \times d}$
- 4: Let $\phi(\mathbf{x}) = \sqrt{2\beta/V} \{\cos(W\mathbf{x} + b), \sin(W\mathbf{x} + b)\}$ and $\Phi = [\phi(\mathbf{x}_1), \dots, \phi(\mathbf{x}_V)]$
- 5: Thompson samples are given by $g_m(\mathbf{x}) = \phi(\mathbf{x})^T (\Phi\Phi^T + \sigma^2 I)^{-1} \Phi^T \mathbf{y}$
- 6: Use a global optimizer to find $g_m^* = \min_{\mathbf{x} \in \mathcal{X}} g_m(\mathbf{x})$
- 7: **end for**
- 8: output: $g_1^*, \dots, g_M^*$

Appendix 2: brief introduction of GEI

Recalling Eq. (3), the EI acquisition function tries to achieve a balance in between the exploitation and exploration. However, it is commonly believed that EI has a tendency to over-exploit. Then, in order to address the “over-exploitation” issue and to achieve a dynamic balance in between exploitation and exploration in the iteration process, GEI (Ponweiser et al. 2008) is proposed which extends the formula of the classical EI as follows:

$$\begin{aligned}
 GEI(\mathbf{x}) &= E(I(\mathbf{x})^g) = \sigma^g \sum_{k=0}^g (-1)^k \binom{g}{k} z^{g-k} T_k \\
 z &= (y_{\text{PBS}} - \hat{y}(\mathbf{x})) / \sigma(\mathbf{x}) \\
 T_0 &= \Phi(z) \\
 T_1 &= -\phi(z) \\
 T_k &= -z^{k-1} \phi(z) + (k-1) T_{k-2}
 \end{aligned} \tag{26}$$

where, g is the global factor that can be set as $0, 1, 2 \dots n$. When $g = 0$ and $g = 1$, GEI is degenerated to the function of probability of improvement (PI) and EI, respectively. With the increase of g , GEI will put much greater weight on the prediction uncertainty $\sigma(\mathbf{x})$ to enhance exploration of the optimization space. Further, inspired by the strategy of simulated annealing (van Laarhoven and Aarts 1987), g is updated dynamically at different stages of the optimization process, as shown in Table 3.

The potential of GEI to improve the performance of EI lies in the following fact. That is, the exploration of optimization space is enhanced at the early stage using large g , which can be helpful to address the “over-exploitation” issue when $z > 0$. Nevertheless, adjusting g in GEI cannot change the value of z . In other words, GEI cannot resolve the issue of “over-exploitation” when it occurs with $z \leq 0$. Moreover, the iterations with the empirical settings of g in Table 3 may even make the algorithm get into the situation of $z \leq 0$ more easily for higher-dimensional problems.

Appendix 3: brief introduction of PEI

As noted before, the pseudo expected improvement (PEI) (Zhan et al. 2017) is originally proposed to enable parallel BO, which is shown to be also effective to enhance the exploration of EI. More specially, an influence function (abbreviated as IF) is proposed to query samples other than $\max EI$. The expressions of IF and PEI are formulated as:

$$IF(\mathbf{x}, \mathbf{x}^*) = 1 - \exp\left(-\sum_{i=1}^d \|x_i - x_i^*\|^2 / 2l_i^2\right) \tag{27}$$

$$PEI(\mathbf{x}, \mathbf{x}^*) = EI(\mathbf{x}) \times IF(\mathbf{x}, \mathbf{x}^*) \tag{28}$$

where, $\mathbf{x}^*$ can be $\max EI$ or other selected sample candidate using PEI, d is the problem dimension, and l_i is the bumpiness parameter of the squared exponential kernel along the dimension i .

In Eq. (27), the IF value is zero if $\mathbf{x} = \mathbf{x}^*$, and approaches to one as $\|\mathbf{x} - \mathbf{x}^*\| \rightarrow \infty$. Hence, when maximizing PEI to choose the next sample candidate to query, $\mathbf{x}^*$ is taken as a landmark to facilitate the algorithm into exploring the promising areas other than the vicinity of $\mathbf{x}^*$. However, neither the value of z nor the weight of $\sigma(\mathbf{x})$ is adjusted in Eq. (28). Therefore, PEI may do help to resolve the issues of “over-exploration” and “over-exploitation”, but it cannot address the issues directly.

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Declarations

Conflicts of interest The authors declare that they have no conflict of interest.

Replication of results The code for replication of results can be found through the website link: <https://github.com/ZhendongGithub/CR-EI>.

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